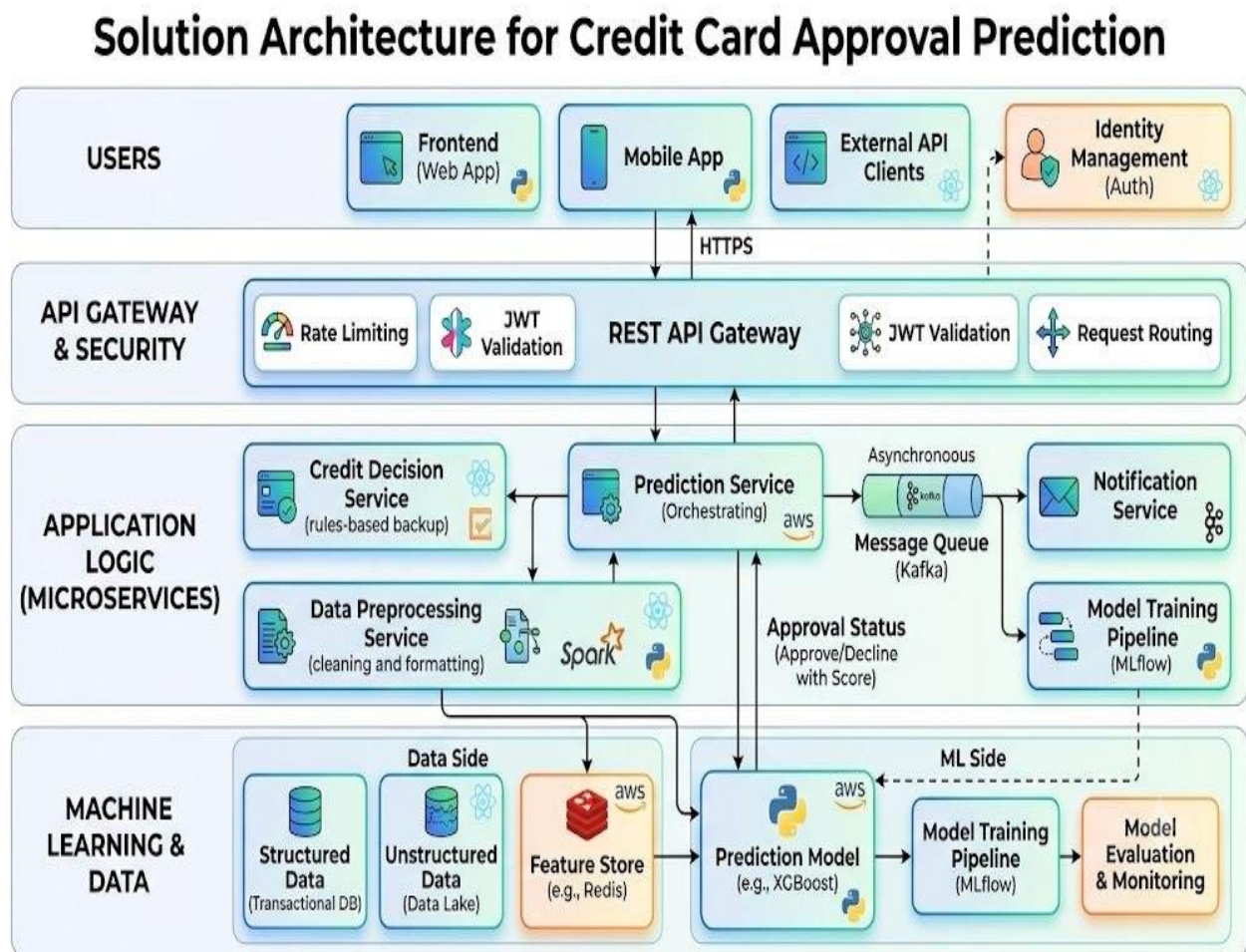


Solution Architecture

Date	26 June 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Layer	Component Name	Description / Role in Architecture	Technologies Used
Users / Presentation	<i>Frontend (Web App) & Mobile App</i>	<i>Provides user-facing interfaces where bank analysts or applicants submit credit card profiles and view real-time approval decisions and performance charts.</i>	<i>HTML5, CSS3, JavaScript, Python (Flask)</i>
	<i>External API Clients</i>	<i>Enables third-party systems or banking applications to interact programmatically with the prediction services.</i>	<i>REST API protocols, React / JSON</i>
	<i>Identity Management (Auth)</i>	<i>Handles user authentication and role-based permissions (e.g., Analyst vs. Compliance Officer) before allowing access to internal logic.</i>	<i>OAuth2 / OpenID Connect</i>
API Gateway & Security	<i>REST API Gateway</i>	<i>Acts as the single entry point for all client requests, managing security rules, Rate Limiting, and Request Routing to appropriate microservices.</i>	<i>Nginx / AWS API Gateway</i>

	<i>JWT Validation</i>	<i>Inspects and validates stateless JSON Web Tokens to protect downstream microservices from unauthenticated traffic.</i>	<i>JWT (JSON Web Tokens)</i>
Application Logic (Microservices)	<i>Prediction Service (Orchestrating)</i>	<i>The core coordinator service that receives user payloads, triggers preprocessing, queries the ML model, and pushes results downstream.</i>	<i>Python, Flask, AWS</i>
	<i>Credit Decision Service</i>	<i>Provides a rules-based fallback engine to evaluate basic credit baseline thresholds if the ML service is unreachable.</i>	<i>Python / Business Rules Engine</i>
	<i>Data Preprocessing Service</i>	<i>Cleans, normalizes, and encodes incoming raw applicant data fields to match the exact mathematical formats required by the ML model.</i>	<i>Apache Spark, Python</i>
	<i>Message Queue (Kafka)</i>	<i>Handles asynchronous communication by streaming transaction events and prediction logs reliably between decoupling layers.</i>	<i>Kafka consumer, Email/SMS API</i>
	<i>Notification Service</i>	<i>Pulls processed events from the message queue to send automated alerts, emails, or system logs to the user/analyst</i>	<i>Kafka consumer, Email/SMS API</i>

		<i>regarding the credit card decision.</i>	
	<i>Model Training Pipeline</i>	<i>Tracks, registers, and packages updated model versions automatically away from the active prediction flow.</i>	<i>MLflow, Python</i>
Machine Learning & Data	<i>Structured & Unstructured Data</i>	<i>Hosts historical applicant credit records, demographic profiles, and high-velocity transactional databases.</i>	<i>Relational DB (SQL) / Data Lake</i>
	<i>Feature Store</i>	<i>Acts as a low-latency cache to store and serve pre-computed operational machine learning features for rapid inference.</i>	<i>Redis, AWS</i>
	<i>Prediction Model</i>	<i>Loads the active, serialized championship machine learning weights to execute instant scoring arrays on inbound applications.</i>	<i>XGBoost, Python, AWS</i>
	<i>Model Evaluation & Monitoring</i>	<i>Continuously logs inference drift, checks production prediction accuracy, and triggers automated alerts if performance shifts over time.</i>	<i>MLflow / Monitoring Dashboards</i>